

Ella Ghazarian

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EDUCATION

University of Michigan

Bachelor of Science in Data Science | 3.53 GPA

Ann Arbor, MI

Expected May 2029

- **Relevant Coursework:** Intro to Computing (C++), Object-Oriented Programming, Applied Linear Algebra, Intro to Data Science

EXPERIENCE

Microsoft

May 2025 – Aug. 2025

Software Engineering Intern (Explore Program)

Redmond, WA

- Engineered a backend API (C#/.NET, Azure CosmosDB) enabling real-time outage data for an internal dashboard supporting 10,000+ users
- Resolved API routing failures and data model mismatches, enabling transition from mock to production data and improving reliability of internal systems
- Designed fault-tolerant multi-call data retrieval across APIs to prevent cascading request failures and ensure reliable outage data during partial service outages
- Strengthened data fetching by refactoring React Query integration to support multi-identifier queries and maintain 60-day historical data availability during API failures
- Coordinated cross-timezone development (15-hour difference) to sequence pull requests, resolve merge conflicts, and unblock integration across dependent components
- Unblocked release by identifying and fixing critical UI and data issues days before deployment, preventing launch delays and restoring full dashboard functionality

University of North Texas

May 2023 – July 2023

Artificial Intelligence & Machine Learning Intern | Poster Paper | Research Paper

Denton, TX

- Implemented machine learning techniques to achieve a 20% accuracy enhancement, introducing aspect-based sentiment analysis for topic-specific sentiment assessment

PROJECTS

SafetyFirst – JacHacks (UMich) | Github Code | Devpost

April 2026

- Designed route safety analysis system using Python, FastAPI, and Google Routes API to detect unsafe transitions in pedestrian navigation
- Modeled routes as ordered graph segments and implemented traversal logic to identify safety degradation at transition boundaries (e.g., residential → highway ramp)
- Addressed gaps in OpenStreetMap data by building a custom segmentation pipeline for fine-grained route analysis
- Exposed safety insights through API endpoints enabling unsafe segment detection and rerouting recommendations

ProductivePanda | Github Code

May 2024 – Aug. 2024

- Built a mood-based task manager using Python, Flask, and the Google Cloud NLP API to analyze input, reorganize to-do lists, and suggest uplifting tasks
- Implemented a dual-database system with Firestore for mood tracking and SQLAlchemy for task storage, increasing efficiency and modularity
- Improved sentiment analysis by detecting negation and sarcasm, enhancing mood classification accuracy
- Engineered a modular MoodUser system for personalized suggestions, integrating Fernet encryption for privacy and Pytest for unit testing

DilemmaAI – 1st @ UncommonHacks (UChicago) | Github Code | Devpost

March 2024

- Developed a game theory-based application where users design strategies for an AI agent to execute and evaluate outcomes
- Led UI/UX design in Figma, creating an intuitive interface for strategy input and simulation
- Debugged and implemented TailwindCSS components for dynamic, non-colliding visual elements

TECHNICAL SKILLS

Languages: C++, C#, Python, JavaScript, HTML/CSS

Frameworks: React, .NET, Bing Maps SDK, Flask, Reflex, Tailwind CSS

Developer Tools: Git, Azure DevOps, Azure CosmosDB, Google Cloud Platform, VS Code, Visual Studio, Pytest